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LAB 3 – Architectural Analysis and Component Design

Problem Statement #12 – Blood Bank Inventory & Emergency Donor Matcher

1. Comparison of Architectural Styles

Before selecting an architecture for the Blood Bank Inventory & Emergency Donor Matcher, three common architectural styles were considered: Layered, Microservices, and Client-Server.

Selected Architecture

After comparing the three options, Microservices Architecture was selected for this system.

The main reason is that the system has several different functions, such as managing blood inventory, monitoring expiry, processing emergency requests and allocations, matching compatible nearby donors, and sending emergency SMS notifications. These functions can be separated into individual services and can work together when required.

Architecture	Structure	Advantages	Disadvantages	Suitability
Layered Architecture	Presentation → Business → Data	Simple to understand, well organized, and easy to maintain.	Components can become tightly connected, and scaling individual parts can be difficult.	Good
Microservices Architecture	Independent services such as Inventory, Donor Matching, Notification, and Expiry	Each service can be developed, deployed, and scaled independently. Failure in one service is less likely to affect the whole system.	More complex to develop and maintain, and communication between services can cause network delays.	Best
Client-Server Architecture	Clients communicate with a central server	Simple structure, centralized control, and easier data management.	The central server can become a bottleneck or single point of failure.	Moderate

2. Architecture Selection – Microservices

Why Microservices?

The Blood Bank Inventory & Emergency Donor Matcher has multiple functions that can be separated into independent services. This makes Microservices a suitable architecture for the system.

Reason 1 – Independent Services

Services such as the Inventory Management Service, Expiry Monitoring Service, Donor Matching Service, and SMS Notification Service can work independently. For example, expiry checking does not need to directly perform donor notification. This separation makes the system easier to develop and maintain.

Reason 2 – Better Scalability

Different services may have different levels of usage. For example, the Donor Matching and SMS Notification services may receive many requests during a critical blood shortage. With Microservices, these services can be scaled separately without having to scale the entire application.

Security Advantage

Each service can have its own authentication and authorization checks. This helps ensure that only authorized users and services can access sensitive information such as donor details, blood inventory, emergency requests, and allocation records. Transaction controls can also prevent the same blood bag from being allocated to multiple requests.

Performance Advantage

The Donor Matching and SMS Notification Services can handle emergency matching and notification separately from other functions. This allows resources to be focused on these services when there are many emergency requests and helps the system meet the required 30-second donor notification target.

3. Component Identification

The following five components are identified for the Blood Bank Inventory & Emergency Donor Matcher.

1. Emergency Request & Allocation Service

This service manages emergency blood requests and allocation operations.

It handles:

- Emergency blood requests
- Blood availability requests
- Compatible-unit allocation
- Allocation status
- Emergency request tracking

2. Inventory Management Service

This service manages the blood-bank inventory.

It handles:

- Blood units and components
- Blood groups
- Quantities
- Inventory updates
- Available stock

3. Expiry Monitoring Service

This service monitors blood-unit shelf lives.

It handles:

- Expiry-date checking
- Near-expiry identification
- Expired-unit identification
- Shelf-life validation before allocation

4. Donor Matching Service

This service finds compatible eligible donors during critical shortages.

It handles:

- Blood-group compatibility
- Donor eligibility
- 10 km radius matching
- Matching results

5. SMS Notification Service

This service sends emergency alerts to matching donors.

It handles:

- Emergency SMS alerts
- Donor notification status
- Notification delivery requests
- Recording notification results

4. Provided and Required Interfaces

The main interfaces are listed below. For the UML component diagram, these interfaces can be represented using the standard UML lollipop (provided interface) and socket (required interface) notation.

Component	Provided Interface	Required Interface
Emergency Request & Allocation Service	Emergency API, Allocation API	Inventory API, Donor Match API, Expiry API
Inventory Management Service	Inventory API	Expiry API
Expiry Monitoring Service	Expiry API	—
Donor Matching Service	Donor Match API	Inventory API, Notification API
SMS Notification Service	Notification API	—

5. Recommended Component Connections

Connection 1 – Emergency Request & Allocation → Inventory Management

Interface: Inventory API

Purpose: Used to check real-time availability of compatible blood units and update allocation-related inventory information.

Connection 2 – Inventory Management → Expiry Monitoring

Interface: Expiry API

Purpose: Used to check shelf life and identify expired or approaching-expiry blood units.

Connection 3 – Emergency Request & Allocation → Donor Matching

Interface: Donor Match API

Purpose: Used to initiate compatible donor matching when sufficient blood inventory is unavailable.

Connection 4 – Donor Matching → SMS Notification

Interface: Notification API

Purpose: Used to send emergency SMS alerts to eligible compatible donors within the required 10 km radius.

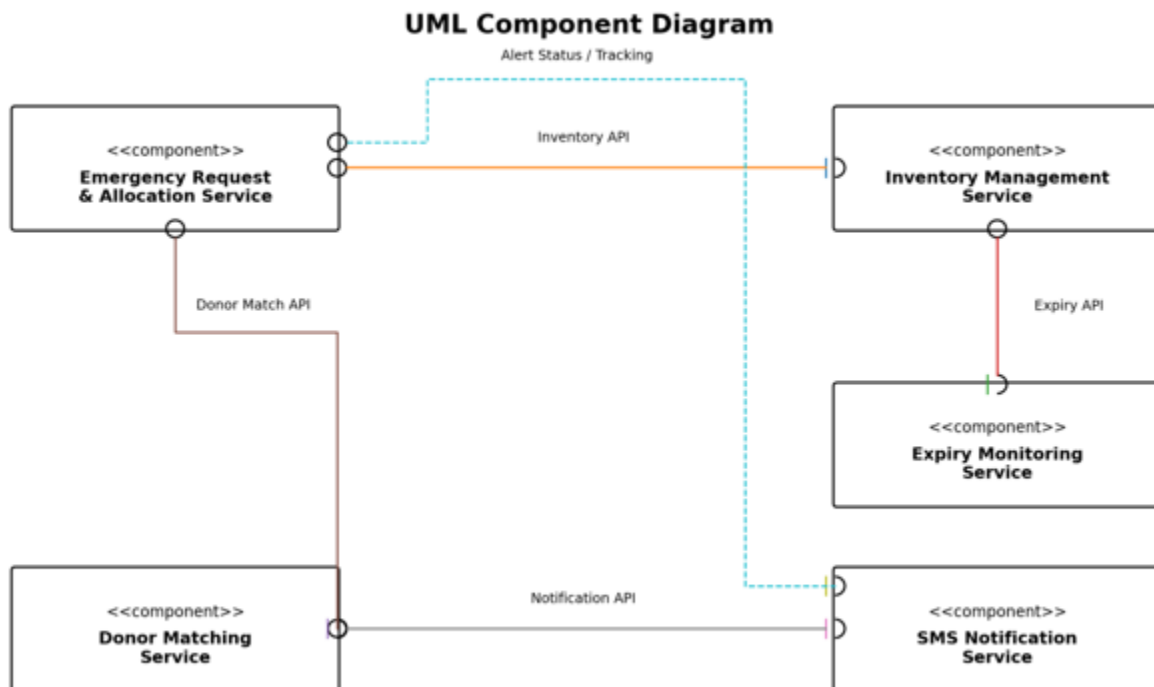
Connection 5 – Emergency Request & Allocation → Expiry Monitoring

Interface: Expiry API

Purpose: Used to validate shelf-life information before a blood unit is allocated.

6. UML Component Diagram

The diagram below represents the five identified components and their provided/required interfaces, following the same overall presentation style as the reference Lab 3 submission.



7. Component Interaction Summary

1. The Emergency Request & Allocation Service receives an emergency blood request and checks the Inventory Management Service.
2. The Inventory Management Service works with the Expiry Monitoring Service to ensure expired or near-expiry units are identified before allocation.
3. If compatible inventory is sufficient, the Emergency Request & Allocation Service performs the required allocation and tracks the emergency request.
4. If sufficient compatible inventory is unavailable, the Emergency Request & Allocation Service invokes the Donor Matching Service.
5. The Donor Matching Service identifies eligible compatible donors within a 10 km radius and invokes the SMS Notification Service.
6. The SMS Notification Service sends emergency alerts and the notification result is tracked as part of the emergency request.